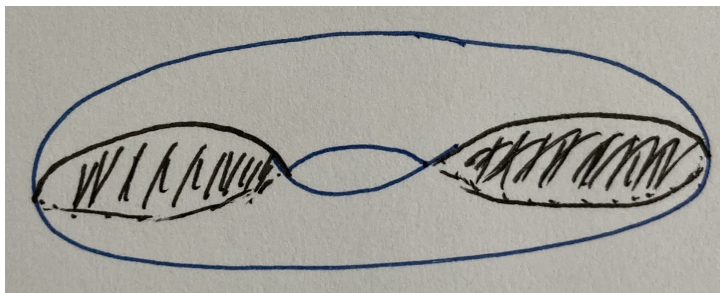


Ph.D. Examination – Topology
August 2022

Work the following problems and show all work. Support all statements to the best of your ability. Work each problem on a separate sheet of paper.

1. Suppose M is a compact 5-manifold with $H_0(M; \mathbb{Z}) = \mathbb{Z}$, $H_1(M; \mathbb{Z}) = \mathbb{Z}_3$, and $H_2(M; \mathbb{Z}) = \mathbb{Z}$. Is M orientable? What are $H_3(M; \mathbb{Z})$, $H_4(M; \mathbb{Z})$, and $H_5(M; \mathbb{Z})$?
2. Let M be a compact, connected, nonorientable 3-manifold. Prove that $H_1(M; \mathbb{Z})$ is infinite.
3. Let X be a connected metric space with metric d and let $p \in X$. Prove that if $X \setminus \{p\} \neq \emptyset$, then $X \setminus \{p\}$ is not compact.
4. Let Σ_g denote the closed surface of genus g , and let $\Sigma_{g,r}$ denote the surface of genus g with r boundary components (i.e., Σ_g with r discs removed). Which of the surfaces $\Sigma_{g,r}$ can admit a map $f : \Sigma_{g,r} \rightarrow \Sigma_{g,r}$ homotopic to the identity that does not have a fixed point?
5. Let X be the space obtained by filling in two discs in the torus, as shown (the discs lie inside the surface of the torus). Compute $H_\bullet(X; \mathbb{Z})$.



Answer the following with complete definitions, statements, or short proofs.

6. Prove or give a counterexample: If $A \subset X$ is path-connected, then the closure $\overline{A}$ is path-connected.
7. Compute $\chi(\mathbb{R}P^2 \times \mathbb{C}P^3 \times S^4)$
8. Give the definition of a normal topological space. Show that a smooth manifold is a normal space.
9. Prove that a continuous bijection from a compact space to a Hausdorff space is a homeomorphism.
10. Prove that if $m \neq n$, then $\mathbb{R}^m$ is not homeomorphic to $\mathbb{R}^n$.

11. Prove that the torus $T^2 = S^1 \times S^1$ is not homotopy equivalent to $S^1 \vee S^1 \vee S^2$.
12. State the Tietze Extension Theorem.
13. Describe all the connected covering spaces $E \rightarrow \mathbb{R}P^2 \vee \mathbb{R}P^3$.
14. Does the following exact sequence of abelian groups necessarily split? Prove or give a counterexample.

$$0 \rightarrow \mathbb{Z}_2 \rightarrow A \rightarrow \mathbb{Z}_2 \rightarrow 0$$

15. Compute the integral homology of the space $\mathbb{C}P^2 \times \mathbb{R}P^2$.